

IntelliCane

Smart Assistive Cane with Haptic Obstacle Avoidance, Fall Detection, and a Companion App for Blind Users and Caretakers

Team Neural-Nexus | Track A: Embedded IoT System Development | IoTrix 2.0

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Repository: <https://github.com/Neural-Nexus-IoTrix-2-0/intellicane> | Submission Date: 5 October 2026

1. The Problem: Why We Built IntelliCane

Navigating everyday surroundings with a traditional white cane can be dangerous and stressful. While a standard cane detects ground-level obstacles, visually impaired individuals frequently bump into knee-to-head level hazards—such as open car doors, low tree branches, or protruding tables. Worse still, an accidental cane drop or user fall can leave someone injured and isolated without anyone knowing. Commercial electronic smart canes attempt to address these risks, but they often cost upwards of \$500 (LKR 150,000+), depend heavily on active internet connections just to detect obstacles, and constantly beep into the user’s ears, drowning out vital ambient sounds.

As an undergraduate engineering team, we built **IntelliCane** to solve these real-world problems directly. At a bench electronics cost of just LKR 2,800, IntelliCane pairs an instant offline hardware reflex on the cane with an accessible Android companion app that keeps caretakers connected without getting in the user’s way.

2. Proposed Solution: How It Works

We designed IntelliCane around two tightly connected components:

- Cane Safety Reflex (ESP32-C3):** Operates 100% offline with a sub-50 ms response loop. An HC-SR04 ultrasonic sensor scans ahead up to 4 m, driving a handle vibration motor via 200 Hz PWM that kicks in gently at 60 cm and smoothly intensifies as obstacles get closer (down to 10 cm). An MPU6050 IMU detects sudden drops and falls ($> 65^\circ$ tilt or $> 2.5g$ impact). When a fall occurs, the cane silences its local motor and buzzer so the user isn’t startled or panicked, while immediately sending an emergency alert packet over BLE.
- Companion Android App** (`com.example.intellicane`): Connects wirelessly over Bluetooth Low Energy (BLE 5.0 Nordic UART Service) with two tailored modes:
 - Blind User Dashboard:* Clean, accessible status cards, one-tap location sharing, and a collapsible floating terminal window to peek at raw sensor telemetry during testing.
 - Caretaker Mode:* Real-time presence indicators, instant cloud fall alerts synced through Firebase Firestore, and an interactive ESRI/OpenStreetMap (OSMDroid) interface with an instant user-centering FAB (18.5x zoom) that uses the phone’s native GPS—saving cost, weight, and battery on the cane itself.

3. System Architecture & Data Flow

Sensing Layer HC-SR04 (GPIO 0/1 direct) MPU6050 (I2C 4/5)	→	Core Embedded Reflex (Cane) ESP32-C3 SuperMini (160 MHz RISC-V) Non-blocking cooperative scheduler	→	Tactile & Audio Alerts 200 Hz LEDC PWM Motor (GPIO 6) Proximity Tone (Silent on Fall)
↓ <i>Wireless BLE 5.0 (Nordic UART Service: RX 6E400002 / TX 6E400003)</i>				
Mobile Application Android Kotlin App (User Dashboard / Caretaker)	→	Location & Cloud Services Smartphone A-GPS Geolocation Firebase Firestore & Presence Sync	→	Caretaker View Real-time OSMDroid/ESRI Map FAB User Centering & Fall Push

4. Hardware Implementation & Circuit Design

We built our prototype around the compact, stamp-sized **ESP32-C3 SuperMini**:

- Ultrasonic Ranging (HC-SR04):** Triggered with $10\mu s$ pulses on GPIO 0. The Echo pulse connects directly to GPIO 1, which works reliably on the ESP32-C3 bench build and keeps wiring minimal and clean.
- Motion & Fall Sensing (GY-521 MPU6050):** Communicates over hardware I2C (SDA: GPIO 4, SCL: GPIO 5) at 100 kHz. We wrote a custom register-level fallback driver to handle clone chips and auto-recover from bus lockups.
- Tactile Vibration Motor:** 3-pin vibration motor module with an onboard MOSFET driver on GPIO 6, modulated at 200 Hz PWM to provide smooth, gradual distance cues without numbing the user’s hand.
- Audible Warning Buzzer:** Active buzzer on GPIO 7 provides urgent beeps for close obstacles (< 30 cm), but automatically stays silent during falls to avoid distressing the user.

5. Firmware Architecture & Control Logic

Our firmware (`firmware/core-sensing/core-sensing.ino`) runs without blocking `delay()` calls, keeping everything responsive:

- Cooperative Loop Timing:** Ultrasonic ranging @ 20 Hz (50 ms), IMU tilt tracking @ 50 Hz (20 ms), motor PWM updates @ 100 Hz (10 ms), and BLE telemetry broadcast @ 10 Hz (100 ms).
- Proportional Haptic Curve:** The motor PWM duty cycle scales smoothly as obstacle distance d decreases:

$$\text{PWM Duty} = \text{constrain} \left(255 - \frac{d - d_{\min}}{d_{\max} - d_{\min}} \times (255 - \text{PWM}_{\min}), \text{PWM}_{\min}, 255 \right) \quad (d_{\min} = 10\text{cm}, d_{\max} = 60\text{cm})$$

- **Silent Fall Detection & BLE Broadcast:** If tilt exceeds 65° or impact exceeds 2.5g, the cane enters an emergency state, silences local vibration and buzzer sound, and broadcasts FALL.STATE:1 over BLE. Bringing the cane upright (< 30°) sends FALL.STATE:0 and instantly restores normal navigation.

6. Companion Mobile Application (application/)

We built the companion app natively in Kotlin (com.example.intellicane) to bridge IoT sensor data to caretakers:

- **BLE Central Client:** Automatically connects to Intelligent-Cane over Nordic UART Service and decodes live telemetry packets: DIST:...|TILT:...|FALL:...|MOTOR:...|BUZZ:...|BLE:....
- **Blind User Dashboard & Floating Terminal:** Gives blind users high-contrast status cards and location sharing. A collapsible floating terminal button (btnToggleTerminal) lets us inspect live telemetry during field tests without cluttering the screen.
- **Caretaker Map & Emergency Centering:** Uses OSMDroid with ESRI map tiles. Caretakers have a dedicated FAB button (fabUserLocation) to jump straight to the user’s live coordinates at 18.5x zoom.
- **Firebase Cloud Backend:** Syncs state with Cloud Firestore. The moment FALL.STATE:1 arrives, the app immediately triggers emergency cloud synchronization to alert caretakers.

7. Testing What We Built & Verification Results

- **Host Unit Tests** (tests/proximity_feedback_test.cpp): 100% pass across boundary conditions, invalid distances (negative, zero, NaN, ∞), smooth PWM ramping, motor polarity flips, and 32-bit millis() rollover safety.
- **Bench Distance Checks:** HC-SR04 verified against physical tape measurements from 2 cm to 250 cm with ±1.5 cm accuracy. Motor vibration engages deterministically at 59.9 cm and reaches full power at 10 cm.
- **Tilt Verification:** Calibrated with a digital protractor; the fall trigger trips reliably at > 65° and resets within 200 ms once the cane is upright (< 30°).
- **Wireless BLE Performance:** Stable 10 Hz stream to smartphone with < 25 ms latency over a 10 m range. Electronics stayed cool (< 32°C) during continuous vibration testing.

8. Budget & Cost Feasibility

We kept prototype costs extremely low by utilizing the smartphone for GPS and cloud connectivity instead of adding expensive dedicated modules:

Component	Specification / Purpose	Phase 1 Cost (LKR)	Status
ESP32-C3 SuperMini	RISC-V 160 MHz MCU with BLE 5.0 & Native USB-C	1,400	Deployed & Verified
HC-SR04 Transceiver	Forward ultrasonic obstacle ranging (2–400 cm)	450	Deployed & Verified
GY-521 MPU6050	6-Axis Accelerometer / Gyroscope for tilt & fall	600	Deployed & Verified
3-Pin Vibration Motor	Integrated driver module (200 Hz LEDC PWM)	250	Deployed & Verified
Piezo Buzzer & Interconnects	Hazard alert tone & bench wiring jumpers	100	Deployed & Verified
Total Electronics Cost	Safety-Critical Bench Prototype	LKR 2,800	Budget Ceiling: LKR 35,000

9. Known Limitations & How We’re Handling Them

- **Ultrasonic Reflections on Soft Clothes:** Sound waves can scatter on angled soft fabrics beyond 2 m. *Fix:* We plan to add a compact Time-of-Flight (ToF) laser sensor as a secondary channel in our final build.
- **Breadboard Jumpers:** Prototype jumpers can loosen under prolonged mechanical tapping. *Fix:* We are migrating to a custom soldered 2-layer PCB and a 3D-printed handle before the final competition.
- **Phone Battery Dependency:** If the phone battery dies, remote cloud tracking pauses. *Safety Reflex:* The cane’s core obstacle avoidance and local haptics stay 100% functional offline.

10. Roadmap Prior to the Final (17 October 2026)

1. **Hardware Enclosure & Custom PCB:** Fabricate a custom PCB and 3D-print an ergonomic cane handle housing with TPU shock damping.
2. **Automated Caretaker Alert Flow:** Finalize Firebase Cloud Functions to dispatch automated SMS and push notifications upon verified fall events.
3. **Battery Optimization:** Integrate an 18650 Li-ion cell with TP4056 charging circuit and configure ESP32-C3 Light Sleep for > 24 hour operation.

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